Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/03/2006
PI Name:	Silber, Mary Ph.D.		
Project Title:	Competing Instabilities and the Spatio-Temporal Dynamics of Interfacial Wave Patterns		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	60208-3125	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	02/15/2000	**End Date:**	11/30/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	note: extended 1 year		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2364		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The work is theoretical and combines linear stability analysis with a weakly nonlinear treatment of the evolution of instabilities. Techniques from dynamical systems and bifurcation theory are exploited, allowing some general statements to be made about consequences of competing parametric instabilities for pattern formation. Our primary investigation focuses on parametrically forced surface waves. For this problem, some especially simple forcing functions, such as impulsive and square wave forcing, allow analytic progress to be made. In other instances, analytic techniques are combined with numerical ones, leading to specific predictions that may be tested in laboratory experiments. This work is aimed at understanding the role that competing parametric instabilities, on different length scales, play in various pattern-forming fluid systems. Of special interest is the Faraday wave system in which surface waves are excited on the interface between two immiscible fluids that are subjected to vibration. This work will help us better understand the effects of vibration on fluid flows, vibrations that may lead to new instabilities or may actually		

	suppress instabilities. It will also contribute to our general understanding of the nonlinear evolution of competing instabilities in hydrodynamic problems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We have developed a scheme for controlling weakly nonlinear three-wave interactions via parametric forcing. This theoretical work grew out of our efforts to understand experiments on Faraday waves, which are excited on the free surface of a fluid layer that is subjected to a vibration in the vertical direction. In particular, these experiments demonstrated that the frequency content of the periodic forcing function can act as an effective switch between different nonlinear standing wave patterns. By focusing on three-wave interactions in the weak-forcing and weak-damping limit, we were able to exploit spatio-temporal symmetries to determine how the forcing parameters, such as frequencies and relative phases, influence the coupling strength between waves driven on different spatial and temporal scales. This work shows how additional frequency components in a periodic parametric drive can be exploited to make delicate superlattice and quasiperiodic wave patterns more robust in the laboratory. We also used this theoretical framework to understand results we obtained using an idealized impulsive forcing function.
Bibliography Type:	Description: (Last Updated: 03/10/2008)
Articles in Peer-reviewed Journals	Topaz CM, Porter J, Silber M. "Multifrequency control of Faraday wave patterns." Phys Rev E Stat Nonlin Soft Matter Phys. 2004 Dec;70(6 Pt 2):066206. PMID: 15697483 , Dec-2004
Articles in Peer-reviewed Journals	Porter J, Topaz CM, Silber M. "Pattern control via multifrequency parametric forcing." Physical Review Letters 2004 Jul 16;93(3):034502. http://dx.doi.org/10.1103/PhysRevLett.93.034502 , Jul-2004
Articles in Peer-reviewed Journals	Higuera M, Riecke H, Silber M. "Near-resonant steady mode interaction: Periodic, quasi-periodic, and localized patterns." SIAM Journal on Applied Dynamical Systems 2004;3(4):463-502. http://dx.doi.org/10.1137/030600552 , Dec-2004
Articles in Peer-reviewed Journals	Porter J, Silber M. "Resonant triad dynamics in weakly damped Faraday waves with two-frequency forcing." Physica D: Nonlinear Phenomena, 2004 Mar 15;190(1-2):93-114. http://dx.doi.org/10.1016/j.physd.2003.09.044 , Mar-2004